

TEJASHVI RAJ

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EDUCATION

Vellore Institute of Technology

2022 – 2026

Bachelor of Technology - Electronics and Communication Engineering; CGPA: 8.63/10 Bhopal, Madhya Pradesh

SKILLS

Programming Languages : C++, Embedded C, Python

Tools and Platforms : ESP32, Arduino, Raspberry Pi, KiCad, Blynk IoT, LTSpice, Tinkercad, SIMULINK, GitHub

Domains : Embedded Systems, PCB Design, Electronics Hardware, Industrial Automation Fundamentals, IoT, VLSI Design

PROJECTS

DHARA : The Ultimate Aqua-Hydro Manager | [View Project](#)

Embedded Systems, IoT

- Developed an ESP32-based real-time monitoring system integrating TDS, pH, turbidity, and temperature sensors for hydroponic/aquaponic applications with automated pH regulation and cloud analytics
- **Technologies**: ESP32, Arduino Uno, Sensors, Ubidots IoT, C++
- **Role**: Embedded logic design, system integration, sensor calibration, and UART-based communication between microcontrollers for real-time data exchange

Smart Bin: IoT-Based Touchless Dustbin | [View Project](#)

Embedded Systems, IoT

- Designed a smart waste monitoring system using ESP32, ultrasonic sensing, and servo motor control with less than 10 ms alert response, enabling real-time overflow detection, automatic lid operation, and cloud notifications
- **Technologies**: ESP32, Ultrasonic Sensor, Tower Pro MG995 Servo Motor, Blynk IoT, C++
- **Role**: Sensor integration, calibration, real-time system monitoring, and firmware optimization for efficient performance

PCB Design Lab | [View Repository](#)

PCB Design / Embedded Hardware Development

- Designed and developed embedded hardware and PCB projects using KiCad, covering schematic capture, PCB layout, routing, design verification, and manufacturing file generation
- **Technologies**: KiCad, PCB Layout, Schematic Capture, DRC Verification, Gerber Generation
- **Role**: Schematic design, footprint assignment, component placement, routing, and design verification

EXPERIENCE

Project Intern

Nov. 2025 – Dec. 2025

Shitashii Innovations Private Limited

IIT Kanpur, UP

- Worked on PCB design and schematic development using KiCad
- Assisted in Raspberry Pi-based hardware interfacing and embedded system tasks
- Contributed to circuit layout optimization and component-level debugging

CERTIFICATIONS

- **Embedded System Design Certification** (Maven Silicon, 2025) [Link](#)
- **VLSI Design Internship – RISC-V and RTL Fundamentals** (Maven Silicon, 2025) [Link](#)
- **MongoDB Associate Database Administrator** (FacePrep, 2025) [Link](#)
- **PCB Design Using KiCad** (Pantech AI, 2025) [Link](#)

CO-CURRICULAR

Co-Head, Electronics Team, VITroniX Club

Jul. 2024 – Feb. 2025